

Fig. 15: Demonstration of a flexible OLED device

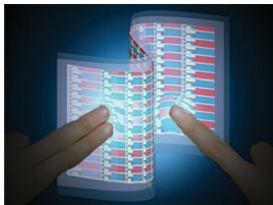


Fig. 16: An artistic illustration of an interactive e-skin device where intensity of the emitted light corresponds to how hard the surface is pressed

manufacturing. Now-a-days the devices are manufactured in such a way that these are embedded in our daily life beyond the limit of separation.

Nano-optoelectromechanical systems (NOEMS) could access energy sources arising at the nanoscale, converting energy from environmental sources, such as ambient noise or electromagnetic radiation, to mechanical vibrations. This route opens up many possibilities where device deformation allows for transformability and new paradigms of user interaction, such as in the Nokia Morph Concept.

There are different ways to harvest or scavenge energy from the surrounding or ambient environment, such as the collection of low-frequency vibrations, heat (via temperature gradients), biomechanical motion and solar energy. Among these, solar energy is perhaps the most promising.

Silicon is by far the most widely used photon absorber and currently dominates the market of PV devices,

with an energy conversion efficiency (η) of up to 25%. Despite significant development over the past decades, the high cost of Si-based solar cells is still a bottleneck for the implementation of solar electricity on a large scale (in the absence of government subsidies). The development of new materials and concepts for PV could be a way to reduce the overall production costs, as well as increase efficiency. The latter is crucial in view of applications in mobile devices that have a limited surface area.

Thin-film solar cells, such as silicon, cadmium telluride (CdTe), copper indium gallium diselenide (CIGS) and thin-film crystalline silicon are termed 'second-generation PVs.' The development of thin-film solar cells has been driven by the potential of cost reduction in manufacturing. An even cheaper and

versatile approach lies in the exploitation of organic photovoltaic (OPV) cells and dye-sensitised solar cells (DSSCs). These can be manufactured economically compared with silicon cells, for example, by roll-to-roll processing, even though they have low η .

In displays and human-machine interactivity

One of the most demanding applications of flexible electronics is in OLED displays. An OLED is a light-emitting diode (LED) in which the emissive electroluminescent layer is a film of organic compound that emits light in response to an electric current. This layer of organic semiconductor is situated between two electrodes. Generally, at least one of these electrodes is transparent. OLEDs are used to create digital displays in such devices as television screens, computer monitors and portable devices like mobile phones and PDAs. A major area of research is the development of white OLED

devices for use in solid-state lighting applications.

There are two main families of OLEDs. Those based on small molecules and those employing polymers. Adding mobile ions to an OLED creates a light-emitting electrochemical cell or LEC, which has a slightly different mode of operation. OLED displays can use either passive-matrix (PMOLED) or active-matrix addressing schemes. Active-matrix OLEDs (AMOLEDs) require a thin-film transistor backplane to switch each individual pixel on or off, but allow for higher resolution and larger display sizes.

An OLED display works without a backlight. Thus, it can display deep black levels and can be thinner and lighter than a liquid crystal display (LCD). In low ambient light conditions, such as a dark room, an OLED screen can achieve a higher contrast ratio than a LCD, whether the LCD uses cold-cathode fluorescent lamps or LED backlight.

e-skin for human-machine integration

You might think that temporary tattoos look cool, but what if they could also collect and transmit information about your heart rate, temperature, muscle contractions or brain waves?

A new flexible electronic circuit promises to do just that, by moving with the skin and staying in place without any adhesive. The research used existing semiconductor technology to imprint integrated circuits onto a thin, flexible silicon film that can be applied directly on the skin. With the interactive e-skin, we could form an elegant system on plastic that can be wrapped around different objects to enable a new form of human-machine interface.

In addition to giving robots a finer sense of touch, the engineers believe that the new e-skin technology could also be used to create things like wallpapers, which double as touchscreen displays and dashboard laminates that allow drivers to adjust electronic controls with the wave of a hand.

The mobile revolution has been